Fault Ride-Through Control for Single-Loop Grid-Forming Inverters: Current Limitation, Compliant Injection, and Synchronization

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***Abstract*—Single-loop voltage-magnitude (SLVM) control offers the grid-forming inverter (GFMI) a simple control structure with robust small-signal stability. During grid faults, however, the SLVM control lacks the inner current loop that would otherwise adjust the modulation voltage to limit the output current, so the current can only be limited by directly shaping this voltage. The shaped voltage, meanwhile, still needs to deliver the grid-code-compliant fault current and keep its phase synchronized with the grid. This paper proposes a fault ride-through (FRT) scheme that fulfills these tightly coupled duties with a current-constrained voltage control and a synchronization-preserving power control, both derived from a single sequence-domain current constraint. The current-constrained voltage control combines a positive-sequence voltage limiter and a negative-sequence voltage reference generator to achieve precise current limitation and grid-code-compliant fault current injection. The synchronization-preserving power control maintains synchronism during FRT via an adaptive active power reference for short-circuit faults and an adaptive voltage feedforward for phase angle jumps and frequency changes. Simulation and down-scaled experimental results verify the proposed scheme under balanced and unbalanced faults, phase angle jumps, and frequency changes.**

***Index Terms*—Grid-forming inverter, current-limiting control, fault ride-through, grid faults, synchronization.**

I. Introduction

DUE TO their inherent frequency and voltage support capabilities, grid-forming (GFM) resources are regarded as indispensable for future power-electronic-dominated power systems [1]. From the perspective of inner-loop implementation, a grid-forming inverter (GFMI) can generally be realized by either a double-loop voltage-current control structure or a single-loop voltage control structure [2].

The double-loop voltage-current control adopts a cascaded architecture, where an outer voltage controller generates the reference for an inner current controller. Within this framework, the GFMI regulates its terminal voltage during normal operation by tracking the prescribed voltage reference, while its output current can be constrained during severe contingencies through a current reference saturation limiter [3], [4].

The single-loop voltage control is commonly implemented as single-loop voltage-magnitude (SLVM) control. Compared with the double-loop structure, SLVM control features a simpler control architecture, in which only the terminal voltage magnitude is regulated, while the phase angle generated by the active power control is directly used for modulation signal synthesis [5]. As a result, the output filter is incorporated into the output impedance in the equivalent voltage source representation of the GFMI [6]. This enlarged impedance alleviates the dynamic coupling between the GFMI and the ac grid, thereby enhancing the small-signal stability [7].

This simplicity, however, becomes a liability during grid faults. Without a current controller, there is no current reference at which the fault current could be readily clamped, so the current can only be limited by directly dictating the modulation voltage. Meanwhile, the inverter-based resource is required to inject positive- and negative-sequence currents prescribed by modern standards [8], [9], and any sustained injection presupposes that the inverter remains synchronized with the grid [10]. Thus, three duties coexist during fault ride-through (FRT) for the SLVM-controlled GFMI: current limitation, grid-code-compliant fault current injection, and synchronization. A double-loop GFMI assigns these duties to separate handles. Its inner current reference is prescribed for the compliant injection and saturated for the limitation, and its outer power loop keeps the synchronization. No such separation exists in the SLVM-controlled GFMI. Instead, the limitation and the injection have to be imposed directly via the modulation voltage, while the synchronization depends on the phase of the same voltage, generated by the active power control. Adjusting this voltage for any one duty thus disturbs the others. The challenge is therefore to fulfill the three coupled duties with the single modulation voltage.

Meeting this challenge imposes several requirements on the control design. First, since no current regulator exists, the current constraint has to be enforced through the modulation voltage itself, using locally measured quantities only. Second, the constraint has to cover both sequences and allocate the limited current headroom between them, so that the overcurrent capability is fully utilized and the sequence injection complies with the standards. Third, the power control has to guarantee that the constrained modulation voltage still keeps the GFMI synchronized during severe disturbances. Existing FRT strategies for the SLVM-controlled GFMI, reviewed below, each fulfill some of these requirements at the expense of another.

For balanced faults, the power reference adjustment in [11] keeps the fault current below the current limit and preserves synchronism by regulating the reference based on the grid voltage. The dependence on the grid voltage gives up the locality demanded by the first requirement. The voltage limiters of [12], [13] translate the current constraint into a magnitude limit

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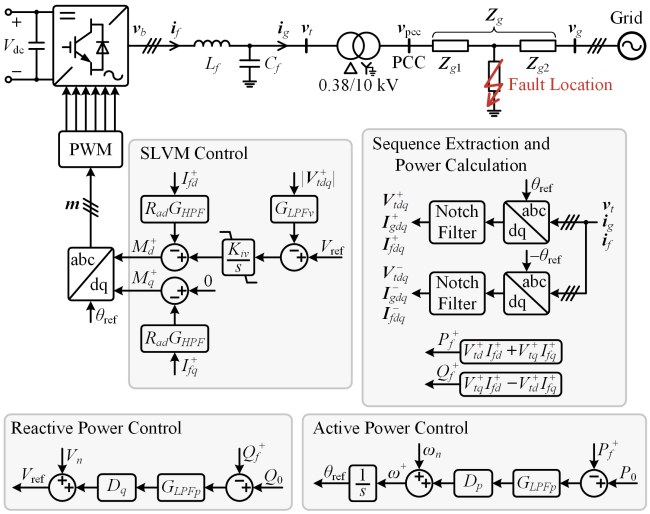


Fig. 1. SLVM-controlled GFMI and its integration into the equivalent ac grid.

and a phase limit on the modulation voltage. The magnitude limit is established locally, but the phase limit usually requires an additional phase-locked loop (PLL) to track the terminal voltage phase, which weakens the applicability in weak grids. Neither strategy covers unbalanced faults.

Unbalanced faults bring in the sequence dimension. The two-stage schemes of [14], [15] bound the transient overcurrent through pulse blocking, but the second stage involves only positive-sequence loops, so the sequence currents are clipped rather than shaped and harmonic distortion arises. The limitation is thus achieved at the price of the compliant injection, a restriction shared by the constraint-aware control of [16]. The sequence voltage limiters of [17] and the voltage-source-sustaining GFM control of [18] shape both sequences and thus address the second requirement. This shaping, however, is obtained with the power loops frozen or left to respond on their own, so the third requirement is sacrificed under phase angle jumps and frequency changes.

For exactly these angle and frequency events, the adaptive fast/slow internal-voltage-source control of [19] combines an adaptive virtual impedance with an enhanced active power control to ride through sags, phase angle jumps, and frequency changes. Nevertheless, it carries no negative-sequence path, so the second requirement reopens under unbalanced faults.

The reviewed strategies thus trade one requirement for another because every corrective action is ultimately exerted on the magnitude and phase of the same modulation voltage. Directly combining these individually effective techniques does not yield reliable FRT either, because the adjustment one makes for its duty overrides what another has shaped for a different duty. What remains missing is a formulation that embeds the sequence current constraints into the SLVM control itself and coordinates the power control around the same constraint, so that balanced and unbalanced faults, phase angle jumps, and frequency changes are addressed within the same formulation. Such a capability is also increasingly demanded by recent GFM specifications [20], [21], [22].

To provide such a formulation, this paper proposes an FRT scheme for the SLVM-controlled GFMI, consisting of a current-constrained voltage control and a synchronization-preserving power control. The two controls comprise four core elements, namely the positive-sequence voltage limiter, the negative-sequence voltage reference generator, the adaptive active power reference, and the adaptive voltage feedforward. All four are derived from one sequence-domain current constraint on the modulation voltage. The voltage elements bound the modulation voltage magnitude in both sequences, and the power elements govern its phase angle, which the constraint leaves free. The two controls thus act on separate parts of the same modulation voltage.

The current-constrained voltage control features:

- Distinct positive- and negative-sequence control loops constructed from the local quantities, achieving precise sequence-domain current limitation.
- Positive- and negative-sequence reactive current support required by modern standards, with full utilization of the available overcurrent capability.

The synchronization-preserving power control features:

- An adaptive active power reference gated by the saturation state of the proposed voltage limiter, maintaining synchronism under balanced and unbalanced faults.
- An adaptive voltage feedforward armed by a terminal voltage threshold drawn from the same constraint circle, enhancing synchronization under phase angle jumps and frequency changes.

The remainder of this paper is organized as follows. Section II introduces the pertinent notations and the SLVM-controlled GFMI. Sections III and IV present the proposed current-constrained voltage control and synchronization-preserving power control, respectively. Section V presents the simulation and down-scaled laboratory results to validate the effectiveness of the proposed method. Section VI concludes this paper.

II. SLVM-Controlled GFMI

A. Notation

Regular italic letters denote scalars, and bold italic letters denote vectors. A bold lowercase letter denotes a three-phase vector in the natural frame, $\boldsymbol{x} = [x_a, x_b, x_c]^{\mathrm{T}}$. A bold uppercase letter with the subscript dq denotes the complex representation of the same quantity in a dq frame, $\boldsymbol{X}_{dq} = X_d + jX_q$, where X_d and X_q are real components and $j = \sqrt{-1}$. The superscripts $+$ and $-$ denote the positive- and negative-sequence components, whose dq frames rotate with the internal phase angle of the GFMI and with its negative, respectively. For a complex quantity X, $|X|$ and $\angle X$ denote its magnitude and angle, so that $X = |X|\angle X$, and a phasor with the named magnitude V and angle θ is written directly as $V\angle\theta$. Angles of dq-frame quantities are measured from the d-axis of their own frame, and phasor angles are measured from the grid voltage phasor. Complex scalars other than dq vectors, e.g., an impedance $Z = R + jX$, are written in regular italics.

B. System Description

Fig. 1 shows the single-line diagram of the SLVM-controlled GFMI and its integrated equivalent ac power grid.

Across the physical circuit, $\boldsymbol{v}_b$, $\boldsymbol{i}_f$, $\boldsymbol{i}_g$, $\boldsymbol{v}_t$, $\boldsymbol{v}_{\rm pcc}$, and $\boldsymbol{v}_g$ denote the bridge voltage, output current, grid current, terminal voltage, point of common coupling (PCC) voltage, and grid voltage, respectively. L_f and C_f represent the output LC filter, a transformer links the terminal to the PCC, and the grid impedance is $Z_g = R_g + jX_g$, with R_g and X_g the grid resistance and reactance, split by the fault location into Z_{g1} and Z_{g2}.

To independently regulate the positive- and negative-sequence components, a sequence-decomposition module (e.g., notch filter) separates them in the dq domain [23], [24]. The GFM control is composed of both active and reactive power control, where droop controls are adopted. The positive-sequence filter-side active and reactive powers, P_f^+ and Q_f^+, are fed back to the droop control, with G_{LPFp} the corresponding low-pass filter (LPF). $P-\omega$ and $Q-V$ droop gains are denoted by D_p and D_q, respectively. The phase angle reference $\theta_{\rm ref}$ generated by the active power control is used for dq transformation, where P_0 is the default active power setpoint and ω_n is the nominal angular frequency. The voltage magnitude reference $V_{\rm ref}$ is regulated by the reactive power control, where Q_0 is the reactive power setpoint and V_n is the nominal voltage magnitude.

The SLVM control is used to generate the dq-axis positive-sequence modulation voltage references, M_d^+ and M_q^+. Specifically, an integrator K_{iv}/s is employed to regulate the terminal voltage magnitude $\left|\boldsymbol{V}_{tdq}^+\right|$ to track $V_{\rm ref}$, with an LPF G_{LPFv} configured to filter out the high-frequency noise. Additionally, the active damping resistor R_{ad} with high-pass filter (HPF) G_{HPF} is added to damp the synchronous resonance introduced by the active power control [19].

III. Current-Constrained Voltage Control

This section develops the current-constrained voltage control, which settles the first two FRT duties, the current limitation and the grid-code-compliant fault current injection, on the magnitude conditions of the modulation voltage.

A. Sequence-Domain Current Constraints

For SLVM-controlled GFMIs, the absence of an inner current controller poses a significant challenge for semiconductor overcurrent protection under grid faults. To safely restrict the output current, the magnitude V_b and phase θ_b of the bridge voltage together with the magnitude V_t and phase θ_t of the terminal voltage need to satisfy the following inequality constraint over the output filter:

$$\frac{|V_b\angle\theta_b - V_t\angle\theta_t|}{X_f} \le I_{\rm lim} \tag{1}$$

where $I_{\rm lim}$ is the current limit and $X_f = \omega_n L_f$.

Existing methods typically translate this constraint into a magnitude limit and a phase limit of the modulation voltage [12], [13], [25]. The magnitude limit can be constructed from locally measured dq quantities, whereas the phase limit bounds the angle between the bridge voltage and the terminal voltage and thus requires the terminal voltage phase as an independent signal, which is typically acquired by a PLL [13], [15], weakening the applicability in weak grids [26]. Moreover, (1) presumes a balanced condition, so it cannot impose a precise sequence current constraint under unbalanced faults.

To overcome these limitations, this section formulates the current constraint directly in the positive- and negative-sequence dq frames defined by the phase angle $\theta_{\rm ref}$ generated by the active power control. In this formulation, only magnitude-level constraints are enforced on the modulation voltage, and the negative-sequence constraint is covered explicitly. The phase degree of freedom is left to the enhanced power control of Section IV, since bounding the angle between the bridge voltage and the terminal voltage does not by itself secure the synchronism, as analyzed in Section III.D.

Both the dq-axis bridge voltage vector $\boldsymbol{V}_{bdq}$ and terminal voltage vector $\boldsymbol{V}_{tdq}$ are decomposed into their corresponding positive- and negative-sequence vectors $\boldsymbol{V}_{bdq}^+$, $\boldsymbol{V}_{bdq}^-$ and $\boldsymbol{V}_{tdq}^+$, $\boldsymbol{V}_{tdq}^-$, with the sequence-domain constraint formulated as

$$\frac{\left|\boldsymbol{V}_{bdq}^+ - \boldsymbol{V}_{tdq}^+\right|}{X_f^+} \le I_{\rm lim}^+, \quad \frac{\left|\boldsymbol{V}_{bdq}^- - \boldsymbol{V}_{tdq}^-\right|}{X_f^-} \le I_{\rm lim}^- \tag{2}$$

where $I_{\rm lim}^+$ and $I_{\rm lim}^-$ denote the prescribed positive- and negative-sequence current limits, and $X_f^+ = X_f^- = X_f$.

The prescribed sequence current limits $I_{\rm lim}^+$ and $I_{\rm lim}^-$ define the inequality constraint in (2). An appropriate allocation of these limits is critical for fully utilizing the current generation capacity of the inverter under fault conditions [3].

The limit $I_{\rm lim}^-$ is typically defined via the K-factor method specified in standards and grid codes [8], [9], [27]:

$$I_{\rm lim}^- = K^- \left|\boldsymbol{V}_{tdq}^-\right| \tag{3}$$

where K^- is configurable in the range of 2 to 6 [9], [27], and $I_{\rm lim}^-$ is additionally capped at $I_{\rm lim}/2$ to retain enough positive-sequence current under severe unbalanced faults [28].

To maximize current utilization under unbalanced faults, at least one phase of the three-phase output currents should reach the phase current limit $I_{\rm lim}$ [8]. Given this, $I_{\rm lim}^+$ is derived from the relationship between the phase current magnitude and the sequence currents:

$$\begin{cases} I_{\rm pha,max} = \sqrt{\left|\boldsymbol{I}_{fdq}^+\right|^2 + \left|\boldsymbol{I}_{fdq}^-\right|^2 + 2I_\Delta} \\ I_\Delta = \max\limits_{\varphi}\left\{\left|\boldsymbol{I}_{fdq}^+\right|\left|\boldsymbol{I}_{fdq}^-\right|\cos\left(\angle\boldsymbol{I}_{fdq}^+ + \angle\boldsymbol{I}_{fdq}^- + \varphi\right)\right\} \end{cases} \tag{4}$$

where $I_{\rm pha,max}$ denotes the maximum magnitude of the phase current and $\varphi \in \{0, -2\pi/3, 2\pi/3\}$.

Imposing $I_{\rm pha,max} = I_{\rm lim}$ on (4), a proper $I_{\rm lim}^+$ that enables full utilization of the inverter's current generation capacity during faults is calculated as:

$$I_{\rm lim}^+ = \sqrt{I_{\rm lim}^2 - \left|\boldsymbol{I}_{fdq}^-\right|^2 - 2I_\Delta} \tag{5}$$

The allocation (3)–(5) follows the established sequence-domain limiting practice [3], [28], and it fixes the radii of the sequence-domain current constraint in (2) on which the subsequent voltage control elements are constructed.

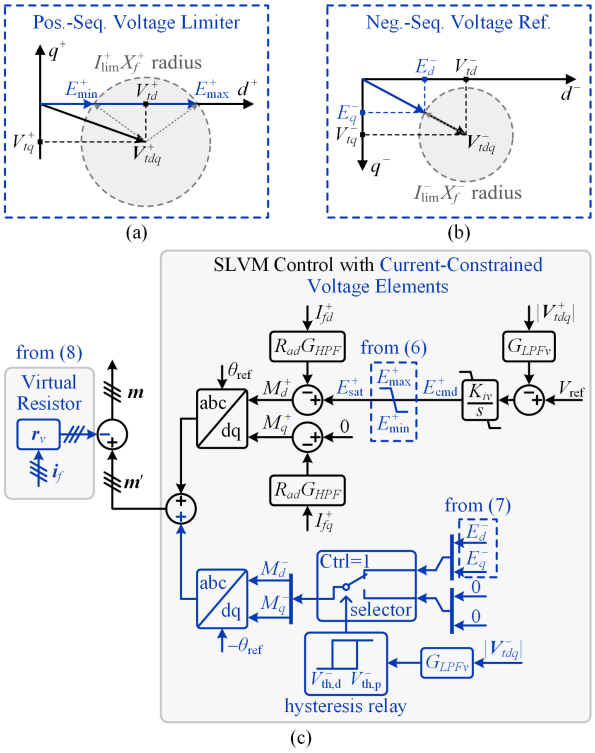


Fig. 2. Proposed current-constrained voltage control. (a) Positive-sequence voltage limiter in the dq^+ plane. (b) Negative-sequence voltage reference generator in the dq^- plane. (c) Implementation within the SLVM control.

B. Sequence-Domain Voltage Control Design

1) Positive-Sequence Voltage Limiter: During the (quasi-) steady state following a grid fault, the positive-sequence dq-axis vectors constrained by (2) are depicted in Fig. 2(a). Geometrically, the positive-sequence constraint defines a circular feasible region in the dq^+ plane, highlighted as the shaded area in Fig. 2(a), centered at $\boldsymbol{V}^+_{tdq}$ with a radius of $I^+_{\rm lim}X^+_f$. The set of points enclosed within this circular boundary represents all viable solutions for $\boldsymbol{V}^+_{bdq}$ satisfying the positive-sequence current constraint.

Under normal operating conditions with negligible control delays, the bridge voltage aligns closely with the modulation voltage, yielding $\boldsymbol{V}^+_{bdq} \approx \boldsymbol{M}^+_{dq}$, where $V^+_{bq} = M^+_q = 0$.

To preserve the voltage source characteristics of the GFMI during severe disturbances, it is desirable to maintain $V^+_{bq} = M^+_q = 0$. Mapping this condition onto the positive-sequence constraint in (2) effectively locks the admissible solution set of $\boldsymbol{V}^+_{bdq}$ to a horizontal line segment along the d^+-axis. The corresponding interval is denoted as $\left[E^+_{\rm min}+j0, E^+_{\rm max}+j0\right]$, with the boundary values derived as

$$\begin{cases} E^+_{\rm max} = V^+_{td} + \sqrt{I^{+2}_{\rm lim}X^{+2}_f - V^{+2}_{tq}} \\ E^+_{\rm min} = V^+_{td} - \sqrt{I^{+2}_{\rm lim}X^{+2}_f - V^{+2}_{tq}} \end{cases} \tag{6}$$

Accordingly, the positive-sequence current constraint in (2) is enforced by a dynamic positive-sequence voltage limiter, bounded by $\left[E^+_{\rm min}, E^+_{\rm max}\right]$, immediately after the integrator output in the SLVM control, as shown in Fig. 2(c).

The feasibility of (6) admits a direct geometric interpretation. The square-root term in (6) stays real-valued as long as $\left|V^+_{tq}\right| \le I^+_{\rm lim}X^+_f$, which is exactly the condition for the current-limiting circle to intersect the d^+-axis. The tangency at $\left|V^+_{tq}\right| = I^+_{\rm lim}X^+_f$ marks the critical case, beyond which no bridge voltage satisfies the current constraint together with $V^+_{bq} = 0$ and the operating point of the limiter is lost. As analyzed in Section IV, this boundary may be approached only under severe phase angle jumps and frequency changes, which significantly enlarge the angle between $\boldsymbol{V}^+_{tdq}$ and the d^+-axis. A severe voltage sag instead reduces $\left|V^+_{tq}\right|$ together with the terminal voltage, so (6) stays feasible. During an infeasible excursion, $E^+_{\rm max}$ and $E^+_{\rm min}$ hold their last feasible values until the adaptive voltage feedforward of Section IV restores the feasibility, and the bounds then resume tracking $\boldsymbol{V}^+_{tdq}$.

Based on the voltage limiter, the positive-sequence fault current is uniquely determined by the restricted $\boldsymbol{V}^+_{bdq}$ and the locally measured $\boldsymbol{V}^+_{tdq}$. Since $E^+_{\rm max}$ and $E^+_{\rm min}$ adapt continuously to $\boldsymbol{V}^+_{tdq}$, they evolve smoothly with grid condition changes. As a result, oscillatory transients and boundary chattering associated with state transitions, including fault occurrence and fault clearing, are effectively suppressed.

2) Negative-Sequence Voltage Reference Generator: According to GFM specifications [29], [30], [31], a GFMI should internally generate a balanced voltage source during near-nominal operating conditions. Accordingly, the negative-sequence dq-axis modulation voltage references M^-_d and M^-_q are set to satisfy $M^-_d = M^-_q = 0$ under normal operations. However, during unbalanced faults, M^-_d and M^-_q should be reformulated to achieve sequence current limitation.

To constrain the negative-sequence current within the prescribed limit $I^-_{\rm lim}$, the negative-sequence dq-axis vectors confined by (2) are illustrated in Fig. 2(b). Similar to the analysis of $\boldsymbol{V}^+_{bdq}$, the negative-sequence constraint defines a circular feasible region in the dq^- plane, centered at $\boldsymbol{V}^-_{tdq}$ with a radius of $I^-_{\rm lim}X^-_f$. The admissible solutions of $\boldsymbol{V}^-_{bdq}$ that satisfy the negative-sequence current constraint in (2) lie within the circular boundary.

In line with recent GFM specifications [21], [29], [32], the GFMI is expected to behave as a balanced voltage source to the extent possible during unbalanced faults. Following this principle, $\left|\boldsymbol{V}^-_{bdq}\right|$ is minimized by aligning $\boldsymbol{V}^-_{bdq}$ (equivalently $\boldsymbol{M}^-_{dq}$) with $\boldsymbol{V}^-_{tdq}$. Consequently, $\boldsymbol{M}^-_{dq}$ is assigned as $\boldsymbol{E}^-_{dq}$ during asymmetric faults, expressed as

$$\boldsymbol{E}^-_{dq} = \left(\left|\boldsymbol{V}^-_{tdq}\right| - I^-_{\rm lim}X^-_f\right)\angle\boldsymbol{V}^-_{tdq} \tag{7}$$

To implement the negative-sequence voltage reference, $\boldsymbol{M}^-_{dq}$ is constrained to $\boldsymbol{E}^-_{dq}$ by a hysteresis relay acting on the filtered $\left|\boldsymbol{V}^-_{tdq}\right|$, which picks up at $V^-_{\rm th,p} = 0.1$ p.u., following the dead band of the grid codes [9], [27], and drops out at $V^-_{\rm th,d} = 0.05$ p.u. Otherwise, $\boldsymbol{M}^-_{dq}$ is set to zero to emulate a fully balanced voltage. Similar to the positive-sequence limiter, this enabling introduces neither chattering nor an abrupt reference change, since the hysteresis band is far wider than the residual ripple of the filtered $\left|\boldsymbol{V}^-_{tdq}\right|$, and the enabled and the disabled references stay close around the threshold, so the smooth boundary evolution is preserved.

3) Transient Overcurrent Suppression: Since the filter inductor dynamics, i.e., the associated differential terms of L_f^+ and L_f^-, are neglected in (2), the constraint mainly applies to steady-state fault conditions. Consequently, the proposed voltage control in (6) and (7) does not inherently suppress the transient overcurrent that may occur during the instants of fault inception and fault clearing. Common methods for mitigating such transient overcurrent include the transient virtual resistor [11] and pulse-width modulation (PWM) pulse blocking [15]. In this study, the proportional transient virtual resistor similar to [11] is implemented upstream of the PWM stage to limit the transient fault current, as shown in Fig. 2(c), where

$$\boldsymbol{r}_v = \begin{cases} 0, & \text{if } |\boldsymbol{i}_f| \le I'_{\lim} \\ K_{rv}\left(|\boldsymbol{i}_f| - I_{\lim}\right), & \text{if } |\boldsymbol{i}_f| > I'_{\lim} \end{cases} \tag{8}$$

where $|\boldsymbol{i}_f|$ takes the magnitude of each phase current and (8) holds phase by phase, so that $\boldsymbol{r}_v$ is the three-phase virtual resistance, and K_{rv} denotes the proportional gain.

The threshold $I'_{\lim}$ is a detection level placed slightly above $I_{\lim}$, so $\boldsymbol{r}_v$ is armed only by the sub-cycle overcurrent at abrupt grid transitions, e.g., fault instants and severe phase angle jumps, and keeps the transient peaks within the short-term capability of the semiconductors. In the (quasi-) steady state, the proposed voltage control settles the phase current at $I_{\lim}$, below $I'_{\lim}$, so the resistor remains inactive and leaves both the current limitation and the reactive current injection required by the standards untouched, while the long-term overcurrent capability is fully utilized.

C. Grid-Code-Compliant Fault Current Injection

A distinctive property of the proposed voltage control is that, beyond the limitation, it also dictates the injected fault current. Once the limiter and the reference generator are active, the sequence currents follow uniquely from the constrained modulation voltage through the output filter. This subsection therefore derives the resulting sequence current contributions and examines them at the grid side, where the standards and grid codes define the injection.

1) Positive Sequence: With the per-unit susceptance B_c^+ of the filter capacitor, the grid-side current is $\boldsymbol{I}_{gdq}^+ = \boldsymbol{I}_{fdq}^+ - jB_c^+\boldsymbol{V}_{tdq}^+$, and the positive-sequence reactive current is

$$I_{Q,g}^+ = \frac{-V_{td}^+ I_{gq}^+ + V_{tq}^+ I_{gd}^+}{\left|\boldsymbol{V}_{tdq}^+\right|} \tag{9}$$

Based on Kirchhoff's voltage law,

$$V_{bd}^+ = V_{td}^+ - I_{fq}^+ X_f^+, \quad V_{bq}^+ = V_{tq}^+ + I_{fd}^+ X_f^+ \tag{10}$$

Under the positive-sequence voltage limiter, the condition $V_{bq}^+ = 0$ holds. Substituting (10) into (9) yields

$$I_{Q,g}^+ = \frac{1}{X_f^+}\left[\frac{V_{bd}^+ V_{td}^+}{\left|\boldsymbol{V}_{tdq}^+\right|} - \left(1 - B_c^+ X_f^+\right)\left|\boldsymbol{V}_{tdq}^+\right|\right] \tag{11}$$

In comparison, the positive-sequence reactive current injection specified in standards and grid codes is [8], [9], [27]

$$I_{Qgc}^+ = K^+\left(1 - \left|\boldsymbol{V}_{tdq}^+\right|\right) \tag{12}$$

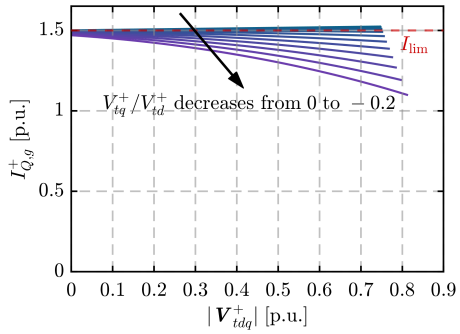


Fig. 3. Positive-sequence reactive current provision of the GFMI under the proposed voltage limiter [calculated using (13)].

where K^+ ranges from 2 to 6 [9], [27].

Equations (11) and (12) present the same voltage-dependent characteristic, in the sense that the positive-sequence reactive current is proportional to the deviation of the terminal voltage magnitude from a reference value. The proportional gain K^+ and the constant reference 1 in (12) are replaced with the gain $1/X_f^+$ defined by the filter reactance and an adaptive reference determined by the voltage limiter, while the capacitor branch enlarges the injection seen by the grid through the factor $\left(1 - B_c^+ X_f^+\right)$. This structural correspondence indicates that the reactive current injection in (11) increases with the depth of the voltage dip, similarly to the behavior described by (12), while remaining bounded by the achievable V_{bd}^+ under the voltage limiter.

In the case of short-circuit faults, the upper limit $E_{\max}^+$ is touched. Substituting $E_{\max}^+$ from (6) into (11) gives

$$I_{Q,g}^+ = \frac{1}{X_f^+}\frac{V_{td}^+\sqrt{I_{\lim}^{+2}X_f^{+2} - V_{tq}^{+2}} - V_{tq}^+}{\left|\boldsymbol{V}_{tdq}^+\right|} + B_c^+\left|\boldsymbol{V}_{tdq}^+\right| \tag{13}$$

According to (13), $I_{Q,g}^+$ under the proposed voltage limiter depends not only on the magnitude $\left|\boldsymbol{V}_{tdq}^+\right|$ but also on the phase angle $\angle\boldsymbol{V}_{tdq}^+$, whereas I_{Qgc}^+ in (12) is solely determined by $\left|\boldsymbol{V}_{tdq}^+\right|$. This difference reflects the voltage source nature of the GFMI. A grid-following inverter can inject reactive current according to the voltage magnitude alone, since its synchronization is handled separately by the PLL. The current of the GFMI is instead the result of the voltage difference across the output impedance, and its split between active and reactive components depends on the angle between the bridge voltage and the terminal voltage, which reduces to $\angle\boldsymbol{V}_{tdq}^+$ with $V_{bq}^+ = 0$ under the limiter. Since the power balance sets this angle to maintain the synchronization, the reactive injection cannot follow the voltage magnitude alone.

The resulting characteristic under the activated limiter is illustrated in Fig. 3 for pre-defined values of $\angle\boldsymbol{V}_{tdq}^+$, where parameters in Table I are used. The angle is taken to be negative because the analysis focuses on the case where the GFMI delivers active power, so $V_{td}^+ > 0$ and $V_{tq}^+ < 0$ according to (10). In Fig. 3, different values of $\angle\boldsymbol{V}_{tdq}^+$ lead to different $\left|\boldsymbol{V}_{tdq}^+\right|$ at which the limiter is activated, so the curves exhibit varying starting points on the right side. $I_{Q,g}^+$ increases

as $\left|\boldsymbol{V}_{tdq}^{+}\right|$ decreases for all the angles, while a more negative $\angle\boldsymbol{V}_{tdq}^{+}$ allocates a larger portion of the current headroom to the active current and thereby yields a lower $I_{Q,g}^{+}$ at the same $\left|\boldsymbol{V}_{tdq}^{+}\right|$. It is also noted that $I_{Q,g}^{+}$ may slightly exceed I_{lim} at the higher $\left|\boldsymbol{V}_{tdq}^{+}\right|$, because the grid-side current contains the capacitor branch contribution $B_c^{+}\left|\boldsymbol{V}_{tdq}^{+}\right|$ in addition to the filter current, while the filter current through the semiconductors remains within I_{lim} at every point of the curves.

2) Negative Sequence: According to IEEE Std 2800-2022 [8], an inverter-based resource is required to inject a negative-sequence current that leads the negative-sequence terminal voltage by 90° to 100°. This angle requirement is critical for establishing a negative-sequence current that supports correct line protection operation and mitigates overvoltage in the healthy phases [28].

When the proposed negative-sequence voltage reference is activated, $\boldsymbol{V}_{bdq}^{-} = \boldsymbol{E}_{dq}^{-}$. Substituting (3) into (7) yields the voltage difference between $\boldsymbol{V}_{bdq}^{-}$ and $\boldsymbol{V}_{tdq}^{-}$ as

$$\boldsymbol{V}_{bdq}^{-} - \boldsymbol{V}_{tdq}^{-} = -K^{-}X_f^{-}\boldsymbol{V}_{tdq}^{-} \tag{14}$$

Based on Kirchhoff's voltage law, $\boldsymbol{V}_{bdq}^{-} - \boldsymbol{V}_{tdq}^{-} = \boldsymbol{I}_{fdq}^{-} jX_f^{-}$. Comparing this relation with (14) gives the relationship between $\boldsymbol{I}_{fdq}^{-}$ and $\boldsymbol{V}_{tdq}^{-}$ as

$$\boldsymbol{I}_{fdq}^{-}/\boldsymbol{V}_{tdq}^{-} = jK^{-} = K^{-}\angle 90^{\circ} \tag{15}$$

Considering the filter capacitor branch with $B_c^{-} = B_c^{+}$, the grid-side negative-sequence current satisfies $\boldsymbol{I}_{gdq}^{-}/\boldsymbol{V}_{tdq}^{-} = (K^{-} - B_c^{-})\angle 90^{\circ}$. The 90° to 100° phase lead required by the standard is therefore preserved at the grid side, and the effective K^{-} seen by the grid is reduced by B_c^{-}, which can be compensated by configuring K^{-} accordingly.

Remark (sensitivity to the filter parameters): In line with the reference SLVM model of [5], the proposed voltage control assumes a purely reactive filter branch with a known reactance. If a parasitic resistance is present and known, it can be included by forming the constraint radius of (6) with the impedance magnitude and rotating the negative-sequence reference of (7) by $\arctan\left(R_f^{-}/X_f^{-}\right)$, which preserves the limiting accuracy and the injection angle. If it is unknown, it raises the impedance magnitude only from X_f to $\sqrt{X_f^2 + R_f^2}$, about 1% for $R_f = 0.15X_f$, and shifts the negative-sequence angle to $90^{\circ} + \arctan\left(R_f^{-}/X_f^{-}\right)$, which stays inside the required 90° to 100° window for $R_f^{-} \le X_f^{-}\tan 10^{\circ}$, a condition that practical filters meet comfortably. If the actual filter reactance slightly differs from the value assumed in the control, the settled current deviates from I_{lim} by a similar margin. An underestimated reactance leaves a corresponding share of the overcurrent capability unused without affecting the injection angle, whereas an overestimated one causes a slight sustained overcurrent that keeps the transient virtual resistor engaged and thereby also moderately shifts the negative-sequence angle. Where the uncertainty is significant, the reactance assumed in the control can be corrected online, e.g., by a proportional-integral regulator on the error between I_{lim} and $I_{\mathrm{pha,max}}$ enabled only while the positive-sequence voltage limiter is saturated, which restores the exact limiting.

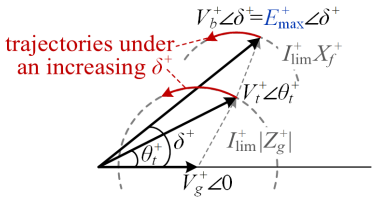


Fig. 4. Phasor diagram when the proposed voltage limiter is activated.

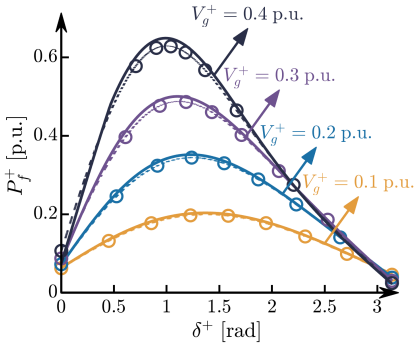


Fig. 5. Power-angle relationships under different grid voltage sags with the proposed positive-sequence voltage limiter activated (solid: calculation results from (16) and (17), dashed with circle markers: simulation results).

D. Transient Stability Under the Proposed Voltage Limiter

With the current limitation and the compliant injection established, the remaining concern of the constrained voltage control is its effect on the transient stability, which can be characterized from the phasor domain. As illustrated in Fig. 4, once the limiter is active, the magnitude of the positive-sequence current phasor is fixed at I_{lim}^{+}. Then, at each power angle δ^{+}, $V_t^{+}\angle\theta_t^{+}$ is determined by the displacement $I_{\mathrm{lim}}^{+}\left|Z_g^{+}\right|$ from $V_g^{+}\angle 0$, and $E_{\max}^{+}\angle\delta^{+}$ is determined by the displacement $I_{\mathrm{lim}}^{+}X_f^{+}$ from $V_t^{+}\angle\theta_t^{+}$. The terminal voltage phasor varies during the transient, but as a dependent quantity fixed by the circuit at each δ^{+} rather than as an independent variable. For a given grid voltage, the delivered active power therefore depends on δ^{+} alone.

With $V_g^{+}\angle 0$ as the reference and the transformer impedance merged into Z_g^{+}, the activated limiter enforces

$$\left|\frac{E_{\max}^{+}\angle\delta^{+} - V_g^{+}\angle 0}{R_g^{+} + j\left(X_f^{+} + X_g^{+}\right)}\right| = I_{\mathrm{lim}}^{+} \tag{16}$$

where the filter capacitor is neglected owing to its negligible effect on transient stability [11].

For a given V_g^{+} and δ^{+}, $E_{\max}^{+}$ is solved from (16), and the delivered active power follows as

$$\begin{aligned} P_f^{+} &= \mathrm{Re}\left\{V_t^{+}\angle\theta_t^{+}\,\mathrm{conj}\left(I_{\mathrm{lim}}^{+}\angle\theta_i^{+}\right)\right\} \\ &\begin{cases} I_{\mathrm{lim}}^{+}\angle\theta_i^{+} = \dfrac{E_{\max}^{+}\angle\delta^{+} - V_g^{+}\angle 0}{R_g^{+} + j\left(X_f^{+} + X_g^{+}\right)} \\ V_t^{+}\angle\theta_t^{+} = I_{\mathrm{lim}}^{+}\angle\theta_i^{+}\left(R_g^{+} + jX_g^{+}\right) + V_g^{+}\angle 0 \end{cases} \end{aligned} \tag{17}$$

where θ_i^{+} is the angle of the positive-sequence filter current.

Fig. 5 compares the characteristics calculated from (16) and (17) with the simulation results for four voltage sag depths. In this verification, the GFMI adopts the original power control of Fig. 1 together with the proposed voltage control of this section, so the effect of the voltage limiter is isolated. The calculated and simulated results agree over the studied range, which validates the power-angle characterization.

The peak of each characteristic in Fig. 5 is the maximum active power deliverable under the corresponding voltage sag. If the active power reference stays below this peak, its intersection with the characteristic defines a fault-on equilibrium. If the reference exceeds the peak, no equilibrium exists and the integral action of the active power control keeps increasing δ^+. A deeper sag lowers the peak and is therefore more likely to lose the fault-on equilibrium.

Notably, the loss of the fault-on equilibrium does not deprive the GFMI of its current-limiting capability. A severe sag reduces $\left|V_{tq}^+\right|$ together with the terminal voltage, so the limiting circle keeps intersecting the d^+-axis and (6) stays feasible. The limiter thus remains saturated while δ^+ grows, and the terminal and bridge voltage phasors slide along the two circles of Fig. 4, as marked by the red trajectories, with the current magnitude pinned at $I_{\lim}^+$. The angle between the two voltages stays bounded along the way, so bounding this angle alone does not secure the synchronism, as noted in Section III.A. A severe sag may therefore cause the loss of synchronism through the vanished equilibrium while the current limitation remains effective, which distinguishes the voltage sags from the phase angle and frequency disturbances analyzed in Section IV.

IV. Synchronization-Preserving Power Control

A. Synchronization Challenge for SLVM-Controlled GFMI

Section III has established the current limitation and the compliant sequence current injection through the constrained modulation voltage. This section addresses the third duty, the synchronization. Based on the active power control in Fig. 1, the angle difference $\Delta\theta_t^+$ between the internal phase angle θ_{ref} and the positive-sequence terminal voltage phase angle θ_t^+ is expressed as

$$\begin{cases} \Delta\theta_t^+ = \theta_{\text{ref}} - \theta_t^+ \\ \theta_{\text{ref}} = \left[\omega_n + D_p\left(P_{\text{ref}} - P_f^+\right) G_{LPFp}\right]/s \\ \theta_t^+ = \omega_t^+/s \end{cases} \tag{18}$$

where ω_t^+ denotes the angular frequency of θ_t^+.

Riding through a severe disturbance requires θ_{ref} to resynchronize with the grid. According to (18), the synchronism is endangered in two ways, either by a sustained power mismatch, which drives θ_{ref} away through the integrator, or by a movement of the grid phase that widens $\Delta\theta_t^+$ and pushes the voltage limiter toward its feasibility boundary.

During severe voltage sags, the deliverable active power collapses while P_{ref} holds its pre-fault value P_0. The sustained mismatch $P_{\text{ref}} - P_f^+$ in (18) keeps θ_{ref} drifting ahead of the grid phase, so the power angle δ^+ diverges. As characterized in Section III.D, the fault-on equilibrium vanishes once P_{ref} exceeds the deliverable peak, whereas the current limitation remains effective.

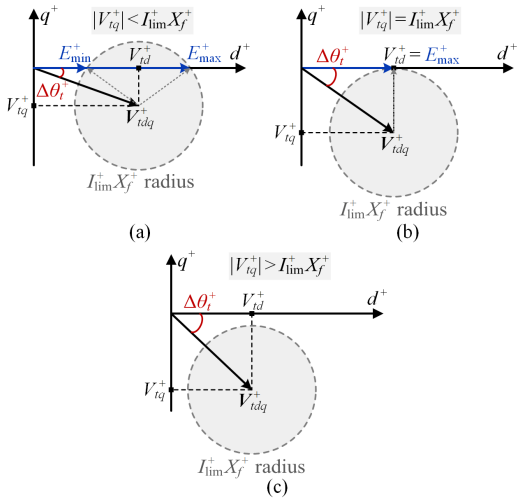


Fig. 6. Positive-sequence current-limiting circle and the d^+-axis. (a) $\left|V_{tq}^+\right| < I_{\lim}^+ X_f^+$: two intersections. (b) $\left|V_{tq}^+\right| = I_{\lim}^+ X_f^+$: tangency. (c) $\left|V_{tq}^+\right| > I_{\lim}^+ X_f^+$: no valid operating point.

During phase angle jumps and frequency changes, the grid phase steps or drifts, while θ_{ref} responds only through the low-bandwidth power control, so the terminal voltage swings away from the d^+-axis and $\Delta\theta_t^+$ widens, resulting in a growing $\left|V_{tq}^+\right|$. With $V_{bq}^+ = 0$, (10) gives $\left|V_{tq}^+\right| = \left|I_{fd}^+\right| X_f^+$, so the growing $\left|V_{tq}^+\right|$ has to be supplied by a growing active current $\left|I_{fd}^+\right|$, which cannot exceed the limited magnitude $I_{\lim}^+$. Geometrically, $\left|V_{tq}^+\right|$ is the distance from the center of the limiting circle to the d^+-axis, so the process runs through the three states of Fig. 6, from the two intersections that define $\left[E_{\min}^+, E_{\max}^+\right]$ in Fig. 6(a), through the tangency at $\left|V_{tq}^+\right| = I_{\lim}^+ X_f^+$ in Fig. 6(b), to the detachment in Fig. 6(c), where the interval vanishes and the equilibrium and the current-limiting feasibility are lost together.

These two mechanisms motivate the proposed synchronization-preserving power control. The adaptive active power reference removes the power mismatch under voltage sags, and the adaptive voltage feedforward drives θ_{ref} to follow θ_t^+ promptly under phase angle jumps and frequency changes. Both are derived from the same sequence-domain current constraint as the voltage control of Section III.

B. Proposed Adaptive Active Power Control

The proposed synchronization-preserving power control is implemented as shown in Fig. 7.

1) Adaptive Active Power Reference Generation: To mitigate the instability caused by grid voltage sags, P_{ref} is adaptively updated to follow P_f^+ measured by the active power control once the proposed positive-sequence voltage limiter is activated. The activation is identified by the output E_{sat}^+ of the limiter departing from its input E_{cmd}^+, so the update reads $P_{\text{ref}} = P_f^+$ when $E_{\text{sat}}^+ \neq E_{\text{cmd}}^+$.

This reference is further bounded so that the sequence reactive current contribution required by Subclause 7.2.2.3.4 of

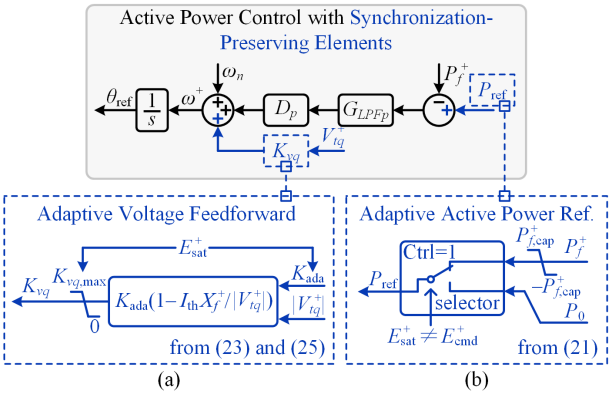


Fig. 7. Proposed synchronization-preserving power control. (a) Adaptive voltage feedforward. (b) Adaptive active power reference generation.

IEEE Std 2800-2022 is fulfilled, namely that the incremental positive-sequence reactive current injection shall not be reduced below the incremental negative-sequence one [8]. Since the negative-sequence current established by the proposed negative-sequence voltage reference is bounded by $I^-_{\rm lim}$ in (3), the reserved positive-sequence reactive current and the corresponding reactive power can be given by

$$I^+_{Q,\rm req} = I^+_{Q,0} + \rho I^-_{\rm lim}, \quad Q^+_{f,\rm req} = \left|\boldsymbol{V}^+_{tdq}\right| I^+_{Q,\rm req} \tag{19}$$

where $I^+_{Q,0}$ is the pre-fault positive-sequence reactive current and ρ is a reserve factor set to 1 in this paper. The reserve is written on $I^-_{\rm lim}$ rather than on a measured negative-sequence reactive current, because it is then available immediately from (3) and is conservative.

While the limiter is activated, $\left|\boldsymbol{I}^+_{fdq}\right| = I^+_{\rm lim}$ fixes the positive-sequence apparent power, so the reserve leaves the active power band of

$$P^+_{f,\rm cap} = \sqrt{\max\left\{0, \left(\left|\boldsymbol{V}^+_{tdq}\right| I^+_{\rm lim}\right)^2 - \left(Q^+_{f,\rm req}\right)^2\right\}} \tag{20}$$

with $-P^+_{f,\rm cap} \le P^+_f \le P^+_{f,\rm cap}$, and capping the active power at its edge forces the positive-sequence reactive current to be at least the reserved value.

The tracking $P_{\rm ref} = P^+_f$ and the band of (20) combine into the final adaptive active power reference:

$$P_{\rm ref} = \begin{cases} P_0, & E^+_{\rm sat} = E^+_{\rm cmd} \\ P^+_{f,\rm cap}, & E^+_{\rm sat} \neq E^+_{\rm cmd},\ P^+_f > P^+_{f,\rm cap} \\ P^+_f, & E^+_{\rm sat} \neq E^+_{\rm cmd},\ \left|P^+_f\right| \le P^+_{f,\rm cap} \\ -P^+_{f,\rm cap}, & E^+_{\rm sat} \neq E^+_{\rm cmd},\ P^+_f < -P^+_{f,\rm cap} \end{cases} \tag{21}$$

Inside the band the reference equals the measured power, so the power angle settles at the value the network imposes. When the measured power reaches the band edge, the reference is clamped there, and the delivered power settles at the deliverable value of the constrained current. The reference returns to P_0 as soon as the limiter leaves saturation, and if $\left|\boldsymbol{V}^+_{tdq}\right|$ drops to zero, $P^+_{f,\rm cap}$ vanishes together with it, so no reference mismatch accumulates under such a condition.

There are also other types of enhanced power reference control, where the reference can be assigned from the residual budget of the rated or limited current [11], [33], [34], reduced in proportion to the voltage dip [35], or held together with only the frozen power loops [17], [36]. The proposed active power reference generation in (19)–(21) differs in two aspects. It is gated by the saturation state of the proposed voltage limiter in both directions, entering at the saturation and returning to P_0 at the de-saturation without any additional threshold or exit rule, and it explicitly reserves the positive-sequence reactive capability required by IEEE Std 2800-2022.

2) Adaptive Voltage Feedforward: During grid phase angle jumps or frequency changes, $\theta_{\rm ref}$ can be adaptively adjusted to rapidly track the variation of θ^+_t, thereby preventing $\Delta\theta^+_t$ from exceeding the predefined feasibility boundary.

According to Fig. 6, the magnitude of $\Delta\theta^+_t$ is positively correlated with that of V^+_{tq}. Thus, when a phase angle jump or frequency deviation changes θ^+_t and drives $\Delta\theta^+_t$ to increase, a feedforward signal based on V^+_{tq} can be introduced to regulate $\theta_{\rm ref}$ [19], [37]. This ensures a stable evolution of $\Delta\theta^+_t$ without divergence, and the modified $\theta_{\rm ref}$ is given by

$$\theta_{\rm ref} = \left[\omega_n + D_p\left(P_{\rm ref} - P^+_f\right) G_{LPFp} + K_{vq} V^+_{tq}\right]/s \tag{22}$$

where K_{vq} denotes the feedforward gain.

Feeding the q-axis terminal voltage back to the synchronization loop as in (22) has been adopted in GFMIs as a virtual damping that enhances the transient stability [38], [39], and its effectiveness has been shown across different grid disturbances [40]. Under the SLVM control, however, the feedforward involves two considerations beyond this damping role. First, under severe phase angle jumps and frequency changes, it has to restore the intersection between the limiting circle and the d^+-axis, namely the existence of the operating point, before the feasibility of (6) is lost. Second, in normal operation, V^+_{tq} settles at $-I^+_{fd}X^+_f$ instead of being regulated toward zero, so a permanently enabled fixed K_{vq} would bias the synchronization loop, while a separate activation logic would introduce additional thresholds to tune. The gain K_{vq} is thus made adaptive and derived from the tangency boundary of the current-limiting circle, so that the feedforward engages only when and as much as the disturbance requires.

The design of K_{vq} should thus satisfy two objectives. First, it remains zero in normal operation and during severe voltage sags, so that the synchronization loop is not biased and the adaptive active power reference is not interfered with. Second, it becomes non-zero during severe phase angle jumps and frequency changes and increases with the disturbance severity to provide sufficient correction.

When the positive-sequence current-limiting circle is tangent to the d^+-axis, $\Delta\theta^+_t$ reaches its feasibility boundary. At this point, the magnitude of V^+_{tq} equals $\left|V^+_{tq}\right|_{\rm bd} = I^+_{\rm lim}X^+_f$. Thus, K_{vq} should become non-zero before $\left|V^+_{tq}\right|$ reaches $\left|V^+_{tq}\right|_{\rm bd}$ to activate the V^+_{tq} feedforward term introduced in (22). Accordingly, K_{vq} is designed as

$$K_{vq} = \max\left\{0,\ K_{\rm ada}\left(1 - I_{\rm th}X^+_f / \left|V^+_{tq}\right|\right)\right\} \tag{23}$$

where K_{ada} is the adaptive gain, and I_{th} is the current threshold satisfying $I_n \leq I_{\mathrm{th}} < I^+_{\mathrm{lim}}$. Here, I_n is the rated current of the GFMI, which is 1 p.u.

The expression of K_{vq} shows that $K_{vq} \neq 0$ if and only if $\left|V^+_{tq}\right| > I_{\mathrm{th}} X^+_f$, and a larger $\left|V^+_{tq}\right|$ yields a greater K_{vq}, which strengthens the feedforward correction and fulfills the second objective. In normal operation, $\left|V^+_{tq}\right| = \left|I^+_{fd}\right| X^+_f$ stays below $I_{\mathrm{th}} X^+_f$, and during severe voltage sags $\left|V^+_{tq}\right|$ decreases together with $\left|\boldsymbol{V}^+_{tdq}\right|$ and remains below $I_{\mathrm{th}} X^+_f$ as well. In both cases $K_{vq} = 0$, which satisfies the first objective.

The design of the adaptive gain K_{ada} in (23) should enable the GFMI to ride through the most severe disturbances among the considered phase angle jumps and frequency changes. Typical cases include negative phase angle jumps or frequency drops, both of which require increased active power injection and thus pose a considerable challenge to the GFMI. Under these conditions, θ^+_t lags further behind θ_{ref}, causing $\Delta\theta^+_t$ to increase. If such disturbances activate the proposed voltage feedforward control, the active power control loop can be mathematically expressed as

$$\Delta\omega^+ = D_p \left(P_{\mathrm{ref}} - P^+_f\right) + K_{\mathrm{ada}} \left(1 - I_{\mathrm{th}} X^+_f / \left|V^+_{tq}\right|\right) V^+_{tq} \tag{24}$$

where $\Delta\omega^+ = \omega^+ - \omega_n$ is the deviation of the internal frequency ω^+ from ω_n, and G_{LPFp} is ignored for simplicity.

Under $V^+_{bq} = 0$, $V^+_{tq} = -I^+_{fd} X^+_f$ follows from (10), and $P^+_f = V^+_{bd} I^+_{fd}$. Hence, V^+_{tq} can be rewritten as $-P^+_f X^+_f / V^+_{bd}$. Substituting this relationship into (24) yields K_{ada} as

$$K_{\mathrm{ada}} = \frac{\Delta\omega^+_u - D_p \left(P_{\mathrm{ref}u} - P^+_{fu}\right)}{\left(-P^+_{fu}/V^+_{bd} + I_{\mathrm{th}}\right) X^+_f} \tag{25}$$

where the subscript u denotes user-defined values.

In (25), V^+_{bd} is the only variable, and it varies with the grid strength. This feature ensures that K_{ada} is adaptive under both weak grid and strong grid conditions. In practice, V^+_{bd} can be replaced by E^+_{sat}. The user-defined parameters are selected according to the most severe disturbance conditions. These conditions include the worst-case frequency deviation, such as $\Delta\omega^+_u = -0.05$ p.u. (corresponding to a 2.5 Hz deviation), the most stringent pre-disturbance operating condition with $P_{\mathrm{ref}u} = 1.0$ p.u., and the desired active power support during disturbances, such as $P^+_{fu} = 1.1$ p.u. Accordingly, K_{ada} can be determined from (25).

In implementation, K_{vq} is capped at the value that (24) assigns under the design condition. Substituting $\Delta\omega^+_u$, $P_{\mathrm{ref}u}$, and P^+_{fu} together with $V^+_{tq,u} = -P^+_{fu} X^+_f / V^+_{bd}$ into (24) and solving for the gain yields $K_{vq,\max} = -\left[\Delta\omega^+_u - D_p \left(P_{\mathrm{ref}u} - P^+_{fu}\right)\right] V^+_{bd} / \left(P^+_{fu} X^+_f\right)$, which coincides with the value that (23) reaches at the design condition. The gain thus grows with the disturbance severity up to the design point $\left|V^+_{tq,u}\right|$ and holds constant beyond it.

3) Complementary Roles of the Two Elements: The two adaptive elements developed above divide the synchronization duty by construction. In a deep voltage sag, the limited current is almost entirely reactive, so I^+_{fd} and with it $V^+_{tq} = -I^+_{fd} X^+_f$ approach zero, the feedforward correction loses its authority, and the sag is carried by the adaptive power reference. Under a phase angle jump or a frequency change, the grid voltage magnitude is essentially unchanged, so the deliverable active power is preserved and no sustained mismatch arises for the reference to remove. The disturbance instead displaces the terminal voltage phase in the control frame, and the resulting growth of $\left|V^+_{tq}\right|$ scales the voltage feedforward correction with the disturbance severity.

TABLE I
SYSTEM PARAMETERS AND CONTROLLER PARAMETERS

Symbol	Description	Value
S_n	Nominal capacity	10 kVA
V_n	Nominal voltage (line-to-line)	380 V
ω_n	Nominal angular frequency	100π rad/s
$Z_{g1} : Z_{g2}$	Fault location	$1:3$
L_f	Filter inductor	0.2 p.u.
C_f	Filter capacitor	0.0315 p.u.
I_{lim}	Inverter phase current limit	1.5 p.u.
I'_{lim}	Transient virtual resistor arming threshold	$1.02 I_{\mathrm{lim}}$
K_{rv}	Transient virtual resistor gain	5
P_0	Active power setpoint	0.7 p.u.
Q_0	Reactive power setpoint	0
D_p	$P-\omega$ droop gain	0.05 p.u.
D_q	$Q-V$ droop gain	0.1 p.u.
ω_{LPFp}	Cutoff frequency of G_{LPFp}	20π rad/s
K_{iv}	Integral gain of SLVM control	10 p.u.
ω_{LPFv}	Cutoff frequency of G_{LPFv}	20π rad/s
R_{ad}	Active damping resistor	0.1 p.u.
ω_{HPF}	Cutoff frequency of G_{HPF}	40π rad/s
K^-	K-factor of neg.-seq. current limit	3
$V^-_{\mathrm{th,p}}/V^-_{\mathrm{th,d}}$	Neg.-seq. voltage thresholds	0.1/0.05 p.u.

V. SIMULATION AND EXPERIMENTAL VALIDATIONS

A. Validation Setup and Performance Metrics

The validation is organized in two layers with distinct roles. The simulation study on the system of Table I establishes the mechanisms and the compliance under a short-circuit ratio (SCR) of 4. It covers the disturbance categories of balanced and unbalanced faults, phase angle jumps, and frequency changes. Moreover, it quantifies the comparison with recent FRT solutions for the SLVM-controlled GFMIs in terms of fault current injection and synchronization. The experimental study on the down-scaled prototype further covers different grid disturbances under SCRs of 13.12 and 2.62. Except for the power and voltage ratings, the GFMI parameters of the two platforms are identical in per unit. The two layers are thus complementary in showing the performance of the proposed control under a wide range of grid strengths (moderate grid: simulation, strong and weak grids: experiment).

The sequence currents and voltages of this section, together with their phase angles, are derived with a one-cycle sliding discrete Fourier transform, and the annotations follow the performance requirements of IEEE Std 2800-2022 [8]:

- The current limitation is annotated with the transient peaks of the phase currents at fault inception and clearing, and with the fault steady-state amplitude of the most stressed phase, both referred to the marked I_{lim}.

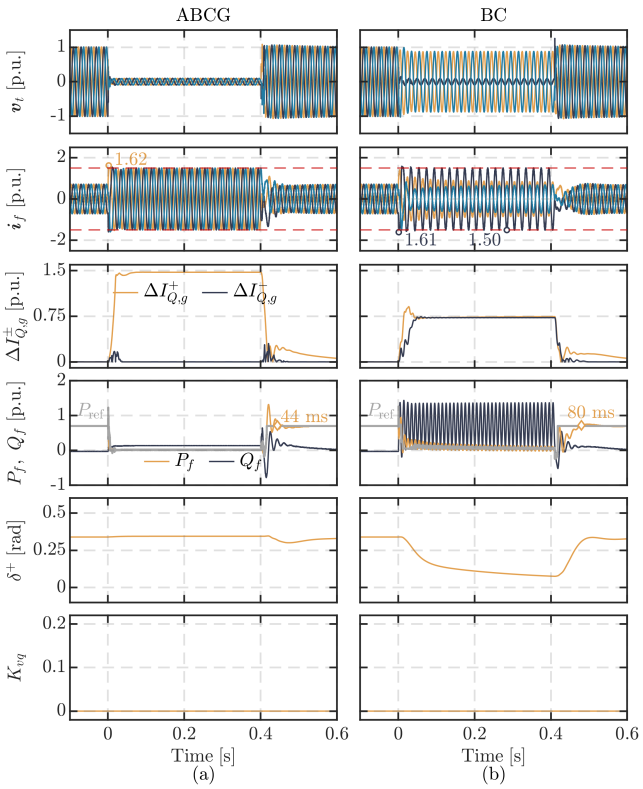


Fig. 8. Simulation results under short-circuit faults, with the fault applied at $t = 0$ s and cleared at $t = 0.4$ s. (a) 3L-G. (b) L-L.

- The unbalanced cases additionally carry the phase angles of the negative-sequence grid current and terminal voltage, shown only during the fault period, so that the lead angle $\Delta\phi^-$ of the current over the voltage, required to lie between $90°$ and $100°$, can be read directly.
- Each incremental sequence reactive current injection $\Delta I^{\pm}_{Q,g}$ is evaluated by the response time $t^{\pm}_r$ to 90% of its fault steady-state value and the settling time $t^{\pm}_s$ into a band of -2.5% to $+10\%$ of I_{lim} around that value, both from the fault inception, listed in Table II.
- The synchronization behavior is evaluated by the recovery time t_{rec}, at which the active power returns to and stays inside $P_0 \pm 5\%$ of the rated power, from the fault clearance for the short-circuit cases and from the inception for the phase angle jumps. The recovery instants are marked by diamonds ($\diamond$) with their values. For the frequency changes, t_{rec} is not defined, since the droop shifts the equilibrium away from P_0.

B. Simulation Studies

MATLAB/Simulink simulations are performed covering different grid disturbances. The benchmark test system is shown in Fig. 1 and the relevant parameters are given in Table I.

1) Short-Circuit Faults: Fig. 8 shows the responses under a balanced and an unbalanced short-circuit fault, applied at the fault location indicated in Fig. 1 at $t = 0$ s and cleared at $t = 0.4$ s. For clarity, 3L-G, LL-G, L-L, and L-G denote the three-line-to-ground, two-line-to-ground, line-to-line, and single-line-to-ground faults, respectively. The plotted 3L-G and L-L cases are examined together with the L-G and LL-G cases in Table II.

TABLE II
FAULT CURRENT INJECTION METRICS OF THE PROPOSED CONTROL

Case	$\Delta\phi^-$ (°)	$\Delta I^+_{Q,g}/\Delta I^-_{Q,g}$ (p.u.)	t^+_r/t^+_s (ms)	t^-_r/t^-_s (ms)
Simulation, short-circuit fault, SCR = 4				
3L-G	–	1.47/0	20/51	–
L-G	92.19	0.74/0.73	21/69	36/37
L-L	92.84	0.73/0.73	13/30	38/45
LL-G	90.91	0.98/0.46	16/59	38/38
Experiment, 3-ph voltage sag				
SCR = 13.12	–	1.45/0	26/31	–
SCR = 2.62	–	1.45/0	23/24	–
Experiment, 2-ph voltage sag				
SCR = 13.12	94.83	0.78/0.67	19/48	45/46
SCR = 2.62	95.68	0.91/0.39	20/36	36/36

Under the 3L-G fault in Fig. 8(a), the inception peak of 1.62 p.u. is damped by the transient virtual resistor within one to two cycles, and the fault steady-state current is then held at I_{lim} for the whole fault period, so the overcurrent capability is fully utilized. The injection is almost purely reactive following the severe balanced fault. Once the voltage limiter saturates, P_{ref} is pulled down with the measured P^+_f, so no power mismatch accumulates in (18). The power angle accordingly stays around its pre-fault value, and the active power recovers to P_0 within 44 ms after the clearance.

Under the L-L fault in Fig. 8(b), the currents are strongly unbalanced while the most stressed phase is held at I_{lim}, and the double-frequency components in the measured powers are the expected consequence of the asymmetric circuit. P_{ref} follows P^+_f toward the deliverable level bounded by $P^+_{f,\text{cap}}$ in (20) and settles at the mean of the oscillating total active power P_f, since the negative-sequence current established by the proposed negative-sequence voltage reference is nearly reactive. The sequence increments settle at $\Delta I^+_{Q,g} = \Delta I^-_{Q,g} = 0.73$ p.u., and the power angle stays steady during the fault with the recovery completed within 80 ms.

Table II examines all four fault types against IEEE Std 2800-2022 [8]. The negative-sequence current leads the negative-sequence terminal voltage by $90.91°$ to $92.84°$, inside the required $90°$ to $100°$ window, as established by the proposed negative-sequence voltage reference generator. The incremental positive-sequence reactive current stays no lower than the negative-sequence one in every case, as enforced by the reactive reserve of the adaptive power reference. Every response time is within the required 2.5 cycles (50 ms) and every settling time within the required 4 cycles (80 ms) as defined in the standard. Moreover, K_{vq} remains at zero throughout the fault cases, so the faults are carried by the adaptive power reference alone.

2) Phase Angle Jumps: Fig. 9 shows the responses to $\pm 30°$ and $\pm 60°$ grid phase angle jumps, applied at $t = 0$ s and $t = 0.4$ s. The current excursions are confined to the jump instants with peaks between 1.57 and 1.69 p.u., and the currents return to the pre-disturbance level in between. It can be seen that K_{vq}

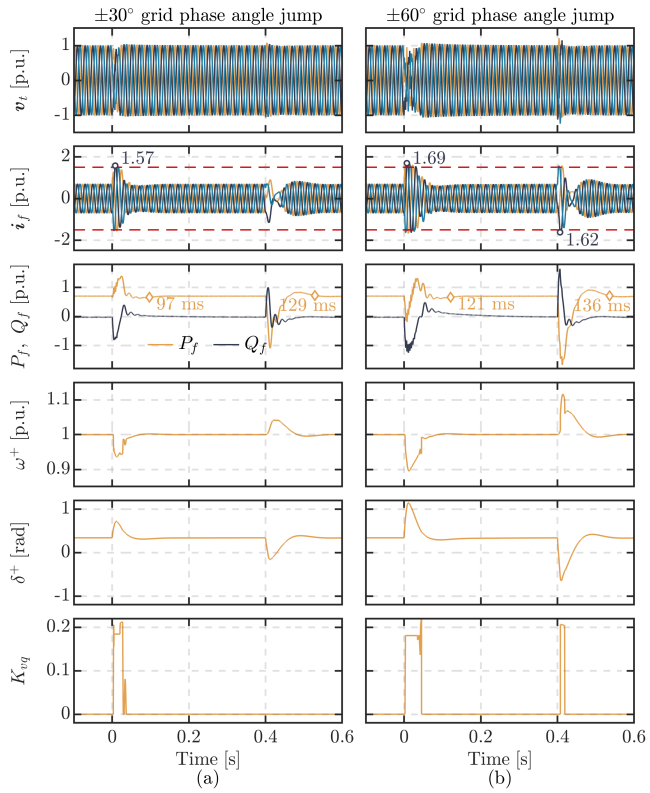


Fig. 9. Simulation results under phase angle jumps, with the negative jump applied at $t = 0$ s and the positive jump applied at $t = 0.4$ s. (a) $\pm 30°$ grid phase angle jump. (b) $\pm 60°$ grid phase angle jump.

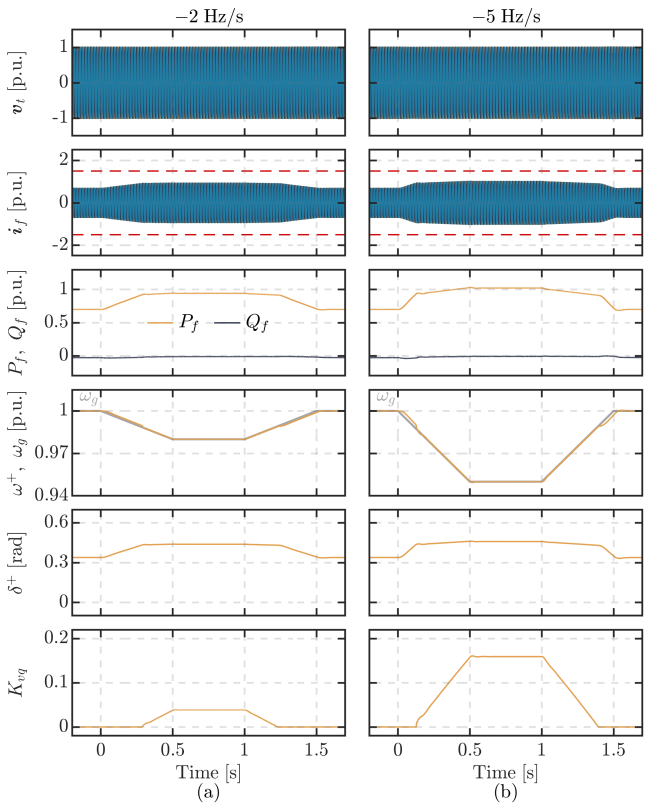


Fig. 10. Simulation results under frequency changes, with the grid frequency ω_g decreasing from $t = 0$ s to $t = 0.5$ s, held constant until $t = 1$ s, and recovering thereafter. (a) 1 Hz grid frequency drop with a RoCoF of -2 Hz/s. (b) 2.5 Hz grid frequency drop with a RoCoF of -5 Hz/s.

rises immediately at each jump, is clamped at $K_{vq,\max}$ during the severe excursions, and returns to zero by itself once the disturbance is traversed.

The negative jumps demand an increased active power injection and thus produce the stronger responses, and the internal frequency dips to 0.94 p.u. at $-30°$ and to 0.9 p.u. at $-60°$. The active power recovers within 97 to 136 ms after the four jumps, and the power angle returns to its pre-disturbance value without divergence even under the $\pm 60°$ jumps, so the adaptive voltage feedforward preserves the synchronization exactly where Section IV.A locates the risk.

3) Frequency Changes: Fig. 10 presents the responses under two rate of change of frequency (RoCoF) events, -2 Hz/s to 49 Hz and -5 Hz/s to 47.5 Hz. As the grid frequency falls, the active power rises through the droop response and provides the frequency support, while the phase currents stay below $I_{\lim}$ and P_f remains bounded by the designed support level $P_{fu}^{+} = 1.1$ p.u. K_{vq} ramps up with the growing frequency deviation, holds during the constant-frequency interval, and ramps back to zero as the frequency recovers, so the feedforward contributes for exactly as long as the event persists. Its plateau grows from about 0.04 under -2 Hz/s to 0.16 under -5 Hz/s, scaling with the disturbance severity as designed. The internal frequency tracks the imposed frequency excursion smoothly, and the GFMI returns to the pre-disturbance operation after the recovery.

Across Figs. 8, 9, and 10, the division of duties established in Section IV.B is directly visible. The short-circuit faults are carried by the adaptive power reference with K_{vq} staying at zero, whereas the phase angle jumps and frequency changes are carried by the adaptive voltage feedforward, with K_{vq} increasing with the disturbance severity.

C. Comparison With Existing FRT Control Schemes

To evaluate the proposed method against existing FRT schemes, three published schemes are considered under identical test conditions. The adaptive fast/slow internal-voltage-source (IVS) control [19] combines adaptive-virtual-impedance-based current limiting with fast-IVS synchronization measures, the two-stage scheme [15] combines PWM-level instantaneous current clipping with active-reactive current limiting, and the scheme of [17] combines sequence internal voltage limiting with frozen power loops. Two disturbances are applied, namely a single-line-to-ground fault applied at $t = 0$ s and cleared at $t = 0.4$ s, and negative grid phase angle jumps of $-30°$ at $t = 0$ s and $-60°$ at $t = 0.4$ s, the negative direction being the demanding one because the inverter is then required to increase its active power injection. The common parameters follow Table I, while the method-specific parameters are selected according to the design procedures reported in the corresponding references. The results are shown in Figs. 11 and 12.

Under the unbalanced fault, the proposed method limits the inception peak to 1.58 p.u. and subsequently maintains the most stressed phase at $I_{\lim}$ with sinusoidal currents. The method in [19] reaches 2.15 p.u. and settles at 1.57 p.u., while the PWM-level clipping of [15] restricts the current

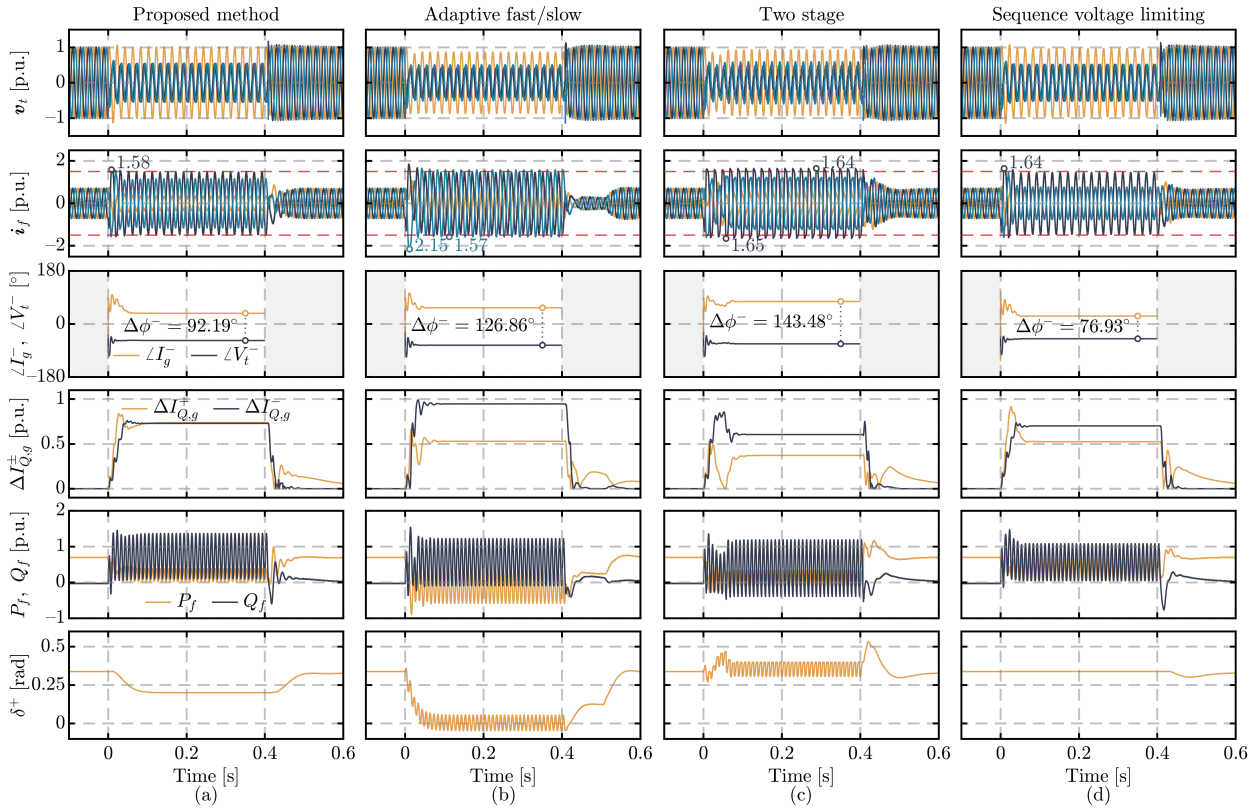


Fig. 11. Comparison under a single-line-to-ground fault. (a) Proposed method. (b) Adaptive fast/slow IVS control [19]. (c) Two-stage current-limiting scheme [15]. (d) Sequence voltage limiting with frozen power loops [17].

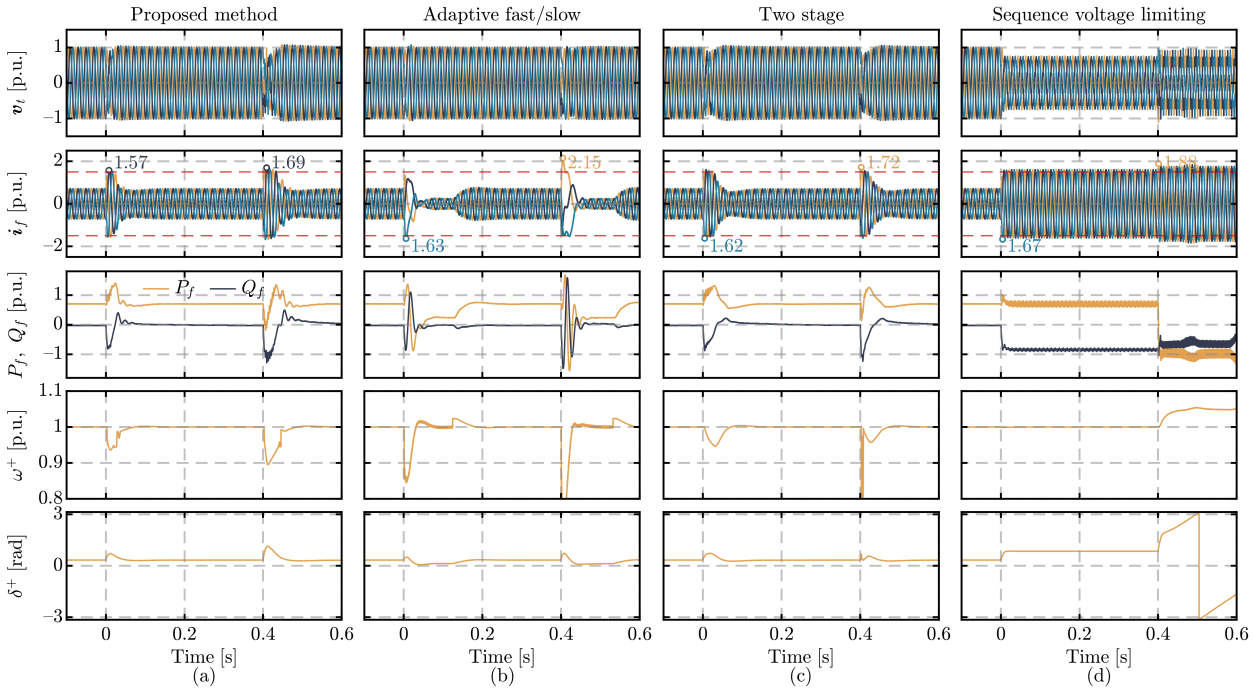


Fig. 12. Comparison under negative grid phase angle jumps of $-30°$ at $t = 0$ s and $-60°$ at $t = 0.4$ s. (a) Proposed method. (b) Adaptive fast/slow IVS control [19]. (c) Two-stage current-limiting scheme [15]. (d) Sequence voltage limiting with frozen power loops [17].

at the cost of waveform distortion. The method in [17] also limits the steady current with sinusoidal waveforms. However, only the proposed method simultaneously establishes $\Delta\phi^- = 92.19°$ within the prescribed $90°$ to $100°$ interval and delivers $\Delta I_{Q,g}^+ = 0.74$ p.u. with $\Delta I_{Q,g}^- = 0.73$ p.u. in accordance with the sequence current requirements of IEEE Std 2800-2022. The corresponding values of the three baselines are $126.86°$, $143.48°$, and $76.93°$ for $\Delta\phi^-$ and $0.53/0.95$, $0.38/0.61$, and $0.53/0.70$ p.u. for $\Delta I_{Q,g}^+/\Delta I_{Q,g}^-$, respectively, none of which meets the corresponding requirement. Moreover, the power

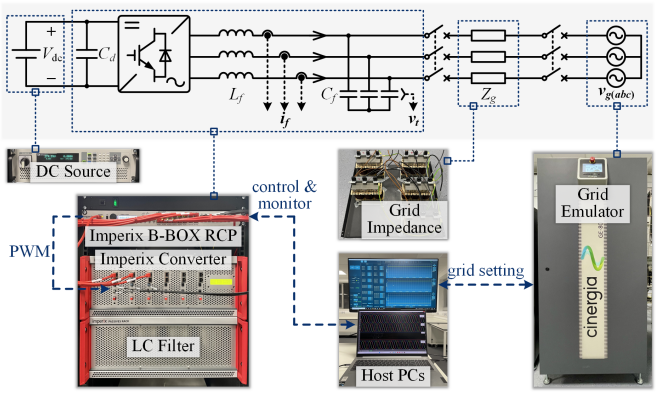


Fig. 13. Experimental setup.

angles stay bounded in all four cases, but those of [15] and [19] carry sustained double-frequency oscillations, since their active power controls are fed with P_f rather than P_f^+, and that of [17] stays constant because its power loops are frozen.

With the mild phase jump of $-30°$, the four methods exhibit comparable inception peaks between 1.57 and 1.67 p.u., yet their power responses already separate. The delivered active power of [19] is suppressed considerably before it recovers, that of [15] recovers with a smaller internal frequency dip, and that of [17] stays depressed with a sustained ripple under its frozen power loops, whereas the proposed method restores P_f rapidly with a brief frequency dip. At the subsequent $-60°$ jump, the peak phase currents are obviously larger for all four methods. The proposed method again recovers the active power rapidly and confines the internal frequency ω^+ to a constrained dip, whereas [19] undergoes large power swings and frequency deviations, while [15] exhibits a sharp frequency excursion down to 0.8 p.u. during transients. The frozen power loops of [17] provide no phase response at all, so its delivered power collapses and ω^+ settles above the nominal value, and the resulting sustained frequency mismatch drives δ^+ to diverge.

The distinguishing advantage of the proposed method is therefore not an isolated improvement in every index, but the simultaneous achievement of phase-current limitation, specified sequence-current support, and synchronization under different grid disturbances.

D. Down-Scaled Laboratory Experimental Test

An experimental setup is developed to validate the proposed control, as shown in Fig. 13. The control of the GFMI is implemented in the Imperix B-Box RCP controller, which drives the Imperix converter modules. A Cinergia grid emulator is employed to reproduce the grid disturbances. The experimental data are collected via the Imperix Cockpit software. In the experiment, $S_n = 1.08$ kVA and $V_n = 104$ V, and the remaining parameters follow the per-unit values of Table I. The tests cover the three-phase and two-phase grid voltage sags, the $\pm 60°$ grid phase angle jumps, and the -5 Hz/s RoCoF event, each recorded under a strong grid with SCR $= 13.12$ and a weak grid with SCR $= 2.62$. The prototype further exposes the control to the sensing errors, the control delay,

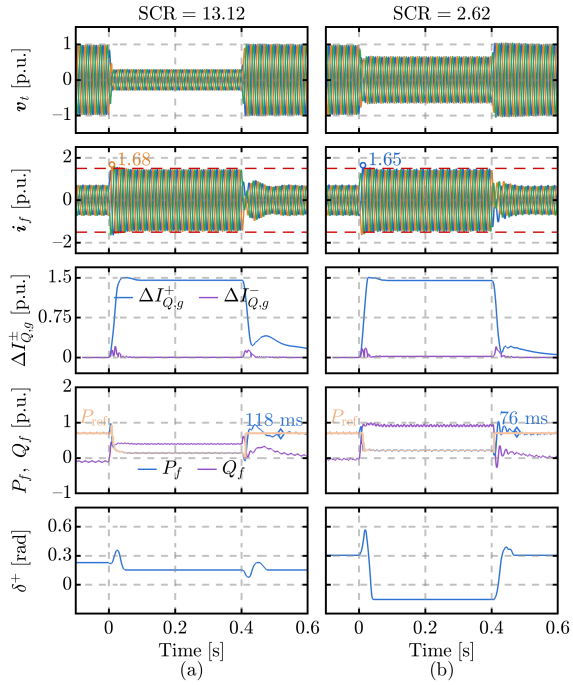


Fig. 14. Experimental results under the three-phase grid voltage sag applied at $t = 0$ s and cleared at $t = 0.4$ s. (a) SCR $= 13.12$. (b) SCR $= 2.62$.

the converter dead time, and the parasitic impedances that the simulation does not contain.

Fig. 14 shows the measured responses under the three-phase grid voltage sag. The phase currents peak at 1.68 p.u. at SCR $= 13.12$ and at 1.65 p.u. at SCR $= 2.62$, and the most stressed phase then stays at $I_{\lim}$ until the clearance. The incremental positive-sequence reactive current settles at 1.45 p.u. while the negative-sequence increment stays at zero under this balanced sag, so the whole current capability is devoted to the positive-sequence voltage support. Once the limiter saturates, the adaptive active power reference again tracks the measured power, so the power angle settles to a constant value during the fault and returns to its pre-fault value after the clearance in both grids, and the fault-on angle differs between the two grids only through the different grid impedance. The active power recovers to P_0 within 118 ms at SCR $= 13.12$ and 76 ms at SCR $= 2.62$.

Fig. 15 shows the two-phase grid voltage sag, where a significant negative-sequence component exists. The inception peaks are 1.66 p.u. and 1.65 p.u., and at least one phase is held at $I_{\lim}$ during the fault, so the current headroom is fully utilized under the unbalanced condition. The injection metrics of both sag cases are collected in the experimental rows of Table II and stay within the same response and settling limits as the simulated cases. In both grids, the negative-sequence current leads the negative-sequence terminal voltage by an angle inside the required $90°$ to $100°$ window, as annotated in Fig. 15, and the incremental positive-sequence reactive current exceeds the negative-sequence one, as listed in Table II. The power angle remains steady during the fault and recovers smoothly after the clearance, with the active power recovered within 85 ms at SCR $= 13.12$ and 71 ms at SCR $= 2.62$.

Fig. 16 shows the $\pm 60°$ grid phase angle jumps. The current

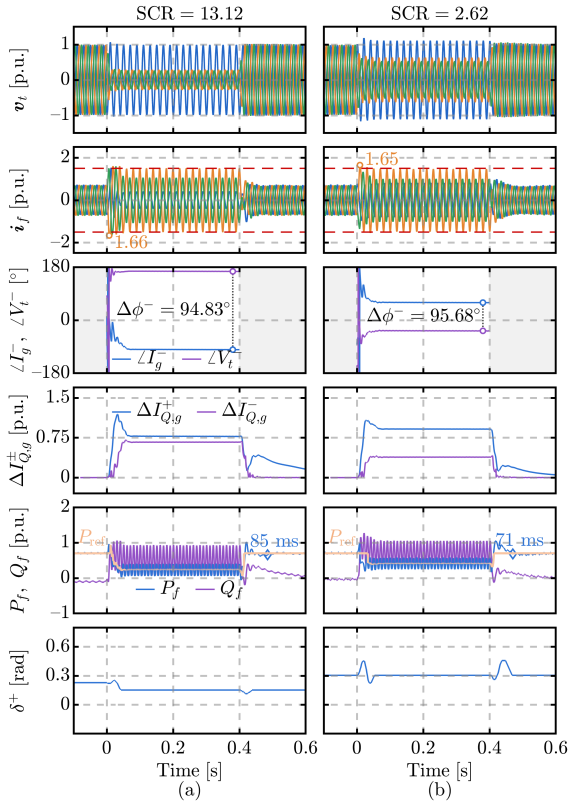


Fig. 15. Experimental results under the two-phase grid voltage sag applied at $t = 0$ s and cleared at $t = 0.4$ s. (a) SCR $= 13.12$. (b) SCR $= 2.62$.

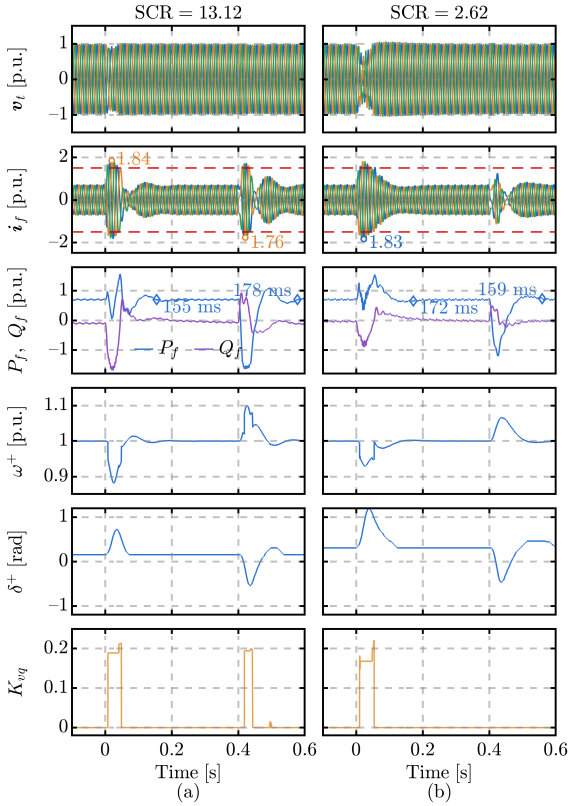


Fig. 16. Experimental results under the $\pm 60°$ grid phase angle jumps, with the negative jump applied at $t = 0$ s and the positive jump applied at $t = 0.4$ s. (a) SCR $= 13.12$. (b) SCR $= 2.62$.

excursions are confined to the jump instants, with peaks of

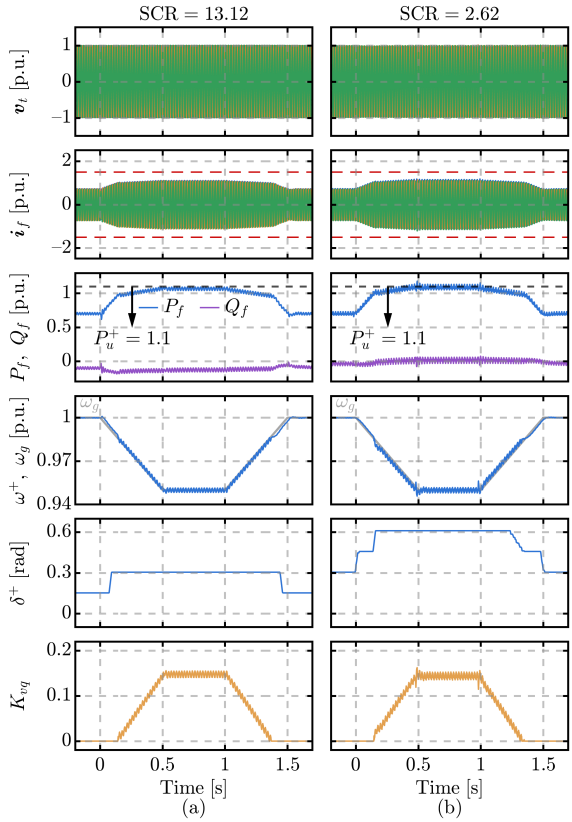


Fig. 17. Experimental results under the -5 Hz/s RoCoF event, with the grid frequency ω_g decreasing from $t = 0$ s to $t = 0.5$ s, held constant until $t = 1$ s, and recovering thereafter. (a) SCR $= 13.12$. (b) SCR $= 2.62$.

1.84 p.u. and 1.76 p.u. at SCR $= 13.12$ and 1.83 p.u. at SCR $= 2.62$, and the currents return to the pre-disturbance level between the two jumps. The feedforward gain K_{vq} steps in only for a short interval around each jump and returns to zero by itself. The active power recovers within 155 to 178 ms after the four jumps. The internal frequency exhibits a brief dip at the negative jump and a brief rise at the positive jump, and the power angle recovers to its pre-disturbance value after each excursion in both grids, so the GFMI rides through the severe phase displacements without losing synchronism.

Fig. 17 shows the -5 Hz/s RoCoF event down to a 2.5 Hz deviation. In both grids, the active power rises with the falling frequency and settles around the designed support level $P_{fu}^+ = 1.1$ p.u. during the constant-frequency interval, while the current stays well below I_{lim}. The internal frequency ω^+ tracks the grid frequency ω_g throughout the ramp, the hold, and the recovery, so the synchronism is retained over the whole event, and K_{vq} engages and disengages with the deviation as in Fig. 10(b). The grid strength shows up mainly in the power angle at which the support is delivered, about twice as large in the weak grid, whereas the delivered power and the gain trajectory are comparable.

Overall, the down-scaled experiments evaluate the current limitation, the sequence current injection, and the synchronization behavior achieved by the proposed method under the hardware non-idealities absent from the simulation and across different grid conditions. These non-idealities also account for the slightly larger transient current peaks and post-disturbance

recovery times compared with the simulation studies.

VI. Conclusion

Without an inner current loop, the SLVM-controlled GFMI carries the current limitation, the fault current injection, and the synchronization all on one modulation voltage. This paper resolves the three coupled duties from one sequence-domain current constraint. The current-constrained voltage control limits the sequence currents and shapes their injection, whose positive-sequence characteristic mirrors the voltage-dependent form and whose negative-sequence angle falls inside the 90° to 100° window, both prescribed by the standards. The synchronization-preserving power control removes the power mismatch under voltage sags through the adaptive power reference and restores the limiting feasibility under angle and frequency events through the adaptive voltage feedforward, both driven by the internal states of the same constraint. Comparative studies against existing methods confirm that the proposed scheme holds all three duties at once under severe grid faults. Down-scaled experiments further validate the scheme across fault scenarios and grid strengths. A rigorous large-signal stability proof of the closed-loop nonlinear system under the proposed scheme remains for future work.

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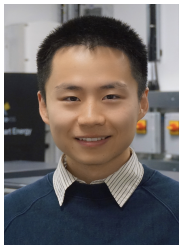
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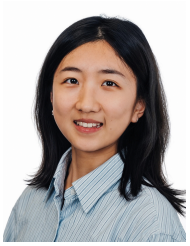
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